

Digital *In Vivo* 3D Atlas of the Antennal Lobe of *Drosophila melanogaster*

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ABSTRACT

As a model for primary olfactory perception, the antennal lobe (AL) of *Drosophila melanogaster* is among the most thoroughly investigated and well-understood neuronal structures. Most studies investigating the functional properties and neuronal wiring of the AL are conducted *in vivo*, although so far the AL morphology has been mainly analyzed *in vitro*. Identifying the morphological subunits of the AL—the olfactory glomeruli—is usually done using *in vitro* AL atlases. However, the dissection and fixation procedure causes not only strong volumetric but also geometrical modifications; the result is unpredictable dislocation and a distortion of the AL glomeruli between the *in vitro* and *in vivo* brains. Hence, to characterize these artifacts, which are caused by *in vitro* processing, and to reliably identify glomeruli for *in vivo* applications, we generated a trans-

genic fly that expresses the red fluorescent protein DsRed directly fused to the presynaptic protein *n*-synaptobrevin, under the control of the pan-neuronal promoter *elav* to label the neuropil in the live animal. Using this fly line, we generated a digital 3D atlas of the live *Drosophila* AL; this atlas, the first of its kind, provides an excellent geometric match for *in vivo* studies. We verified the identity of 63% of AL glomeruli by mapping the projections of 34 GAL4-lines of individual chemosensory receptor genes. Moreover, we characterized the innervation patterns of the two most frequently used GAL4-lines in olfactory research: Orco- and GH146-GAL4. The new *in vivo* AL atlas will be accessible online to the neuroscience community. *J. Comp. Neurol.* 523:530–544, 2015.

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INDEXING TERMS: *Drosophila melanogaster*; olfactory system; glomeruli; *in vivo* neuropil marker; *in vitro* artifacts

The chemical environment consists of a vast and quickly changing array of volatile cues, each of which has different ecological relevance. To orient itself within this complex environment, the vinegar fly *Drosophila melanogaster* possesses an elaborate olfactory system consisting on the periphery of a set of four classes of sensilla present in stereotypical patterns on the third antennal segment and the maxillary palp (Vosshall and Stocker, 2007). Each type of sensillum—antennal basiconic, trichoid, coeloconic, or palp basiconic—houses from one to four olfactory sensory neurons (OSNs). Each OSN expresses a repertoire of one or two chemosensory receptors (CRs); each CR has a unique molecular receptive range (Shanbhag et al., 1999; de Bruyne et al., 2001; Hallem and Carlson, 2006; Silbering et al., 2011). The information gathered on volatile odorants intercepted by the antenna is forwarded to the primary center of the olfactory pathway, the antennal lobe (AL). The *Drosophila* AL represents one of the most comprehensively investigated neuronal structures in nature (Stocker et al., 1990; Masse et al., 2009; Galizia and

Sachse, 2010; Hansson et al., 2010; Wilson, 2013). It consists of discrete neuropil subunits, called olfactory glomeruli, which are conserved among individuals and can be identified by their specific position and size (Laissue et al., 1999). Each glomerulus collects OSN axons of only one type due to the strict convergence of OSNs expressing the same CR type (Gao et al., 2000; Vosshall et al., 2000).

In order to study the odor-coding mechanisms of the AL, a broad spectrum of *in vivo* or *ex vivo* techniques, such as the photolesioning of specified tracts (Liang et al., 2013), the photoactivation of single neurons

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(Datta et al., 2008; Ruta et al., 2010; Caron et al., 2013), patch clamp recordings (Wilson et al., 2004; Chou et al., 2010; Seki et al., 2010), as well as functional imaging (Fiala et al., 2002; Ng et al., 2002; Wang et al., 2003; Silbering et al., 2008; Stökl et al., 2010; Schubert et al., 2014) has been established over the last years. All these techniques demand an adequate atlas for the identification of the anatomical substructures—the glomeruli—so that they can be applied *in vivo*. However, all previous 3D atlases of the *Drosophila* AL have derived from *in vitro* data (Stocker et al., 1983; Laissue et al., 1999; Couto et al., 2005; Endo et al., 2007; Silbering et al., 2011; Tanaka et al., 2012). Since the dissection and fixation procedure causes not only strong volumetric but also geometric modifications (Ma et al., 2008), the result is the unpredictable dislocation and distortion of the AL glomeruli between the *in vitro* and *in vivo* brains. In order to reliably identify glomeruli in the live fly brain, an atlas of the *in vivo* AL is essential.

In this study we present the first digital 3D atlas of the live *Drosophila* AL. To identify glomeruli *in vivo*, we generated a transgenic fly that expresses the red fluorescent protein DsRed directly fused to the presynaptic protein *n*-synaptobrevin, under the control of the pan-neuronal promotor *elav* (Yao et al., 1993) to selectively label the neuropil in the live animal. In addition, we expressed a green fluorescent protein under the control of 34 individual chemosensory receptor genes (Couto et al., 2005; Fishilevich and Vosshall, 2005; Silbering et al., 2011) to confirm the correct identification of most glomeruli by its type of OSN innervation. Once quantified and compared to *in vitro* data, the new *in vivo* data were used to demonstrate nontrivial and unexpected structural deformations and distortions of the majority of glomeruli caused by *in vitro* processing. These results support the need for a digital *in vivo* 3D atlas of the fly AL to find geometric matches for *in vivo* studies. Furthermore, we characterized the glomerular innervation patterns of the two most frequently used GAL4-lines in olfactory research: Orco-GAL4 as well as GH146-GAL4. The new *in vivo* 3D AL atlas will be publicly accessible to the neuroscience community through our website (www.ice.mpg.de/ext/invivoALatlas.html).

MATERIALS AND METHODS

Fly lines

P[END1-2] (*elav n*-synaptobrevin-DsRed 1-2) was generated using a modified pCaST-*elav*-GAL4AD vector (plasmid 15307, Addgene, Cambridge, MA). The GAD domain present in the original vector was excised using NotI and FspAI enzymes; the FspAI recognition site was

located within the DsRed coding sequence. A DNA oligonucleotide containing a modified *n*-synaptobrevin-coding ORF (*n*-syb) (DiAntonio et al., 1993), upstream of a sequence identical to the excised DsRed fragment, and a *Drosophila* Kozak site (caaaATG) and recognition sites for NotI and FspAI were synthesized and inserted into the vector. The *n*-syb contains one silent mutation (C168T) to eliminate an FspAI-recognition site within the fragment. Excision, synthesis, and ligation were performed by MWG Eurofins (Ebersberg, Germany). The resulting plasmid was amplified in *E. coli* (One Shot Top10 *E. coli*, Invitrogen, Eugene, OR) and purified using a Qiagen midi-prep kit (Qiagen, Hilden, Germany). Embryo transformation to generate transgenic lines was performed by Aktogen (Cambridge, UK).

All fly stocks were maintained on conventional cornmeal-agar-molasses medium under a 12-hour light: dark cycle at 25°C. Transgenic lines used were as follows: Gr21a-GAL4 (Scott et al., 2001), Ir40a-GAL4 (RRID:BDSC_41727), Ir41a-GAL4 (RRID:BDSC_41749), Ir75a-GAL4 (RRID:BDSC_41748 (Silbering et al., 2011)), Ir76a-GAL4 (RRID:BDSC_41735 (Benton et al., 2009)), Ir92a-GAL4 (RRID:BDSC_41733 (Abuin et al., 2011)), Or10a-GAL4 (RRID:BDSC_9944), Or13a-GAL4 (RRID:BDSC_23886), Or22a-GAL4 (RRID:BDSC_9951), Or33c-GAL4 (RRID:BDSC_9966), Or35a-GAL4 (RRID:BDSC_9968), Or42a-GAL4 (RRID:BDSC_9969), Or42b-GAL4 (RRID:BDSC_9972), Or43a-GAL4 (RRID:BDSC_9973), Or43b-GAL4 (RRID:BDSC_23895), Or46a-GAL4 (RRID:BDSC_9979), Or47a-GAL4 (RRID:BDSC_9982), Or47b-GAL4 (RRID:BDSC_9984), Or49a-GAL4 (RRID:BDSC_9985), Or56a-GAL4 (RRID:BDSC_23896), Or59b-GAL4 (RRID:BDSC_23897), Or59c-GAL4 (RRID:BDSC_23899), Or67a-GAL4 (RRID:BDSC_23904), Or67c-GAL4 (RRID:BDSC_24856), Or69a-GAL4 (RRID:BDSC_10000), Or71a-GAL4 (RRID:BDSC_23121), Or83c-GAL4 (RRID:BDSC_23132), Or85a-GAL4 (RRID:BDSC_24461), Or85b-GAL4 (RRID:BDSC_23912), Or85d-GAL4 (RRID:BDSC_24148), Or88a-GAL4 (RRID:BDSC_23137), Or92a-GAL4 (RRID:BDSC_23139), Or98a-GAL4 (RRID:BDSC_23141 (Vosshall et al., 2000; Fishilevich and Vosshall, 2005)), Or67d-GAL4 (RRID:BDSC_23906 (Kurtovic et al., 2007)), UAS-GCaMP3.0 (RRID:BDSC_32116 (Tian et al., 2009)), GH146-GAL4 (RRID:BDSC_30026 (Stocker et al., 1997)), Orco-GAL4 (RRID:BDSC_26818 (Larsson et al., 2004)).

In vivo antenna and brain preparation

Between 3 to 6 days after hatching, female flies were anesthetized with CO₂. In order to scan labeled OSN somata on the third segment of the antenna (Fig. 1D), the whole antenna was gently severed using forceps and transferred to a drop of saline with Triton-X on an object holder. The antennae were then gathered

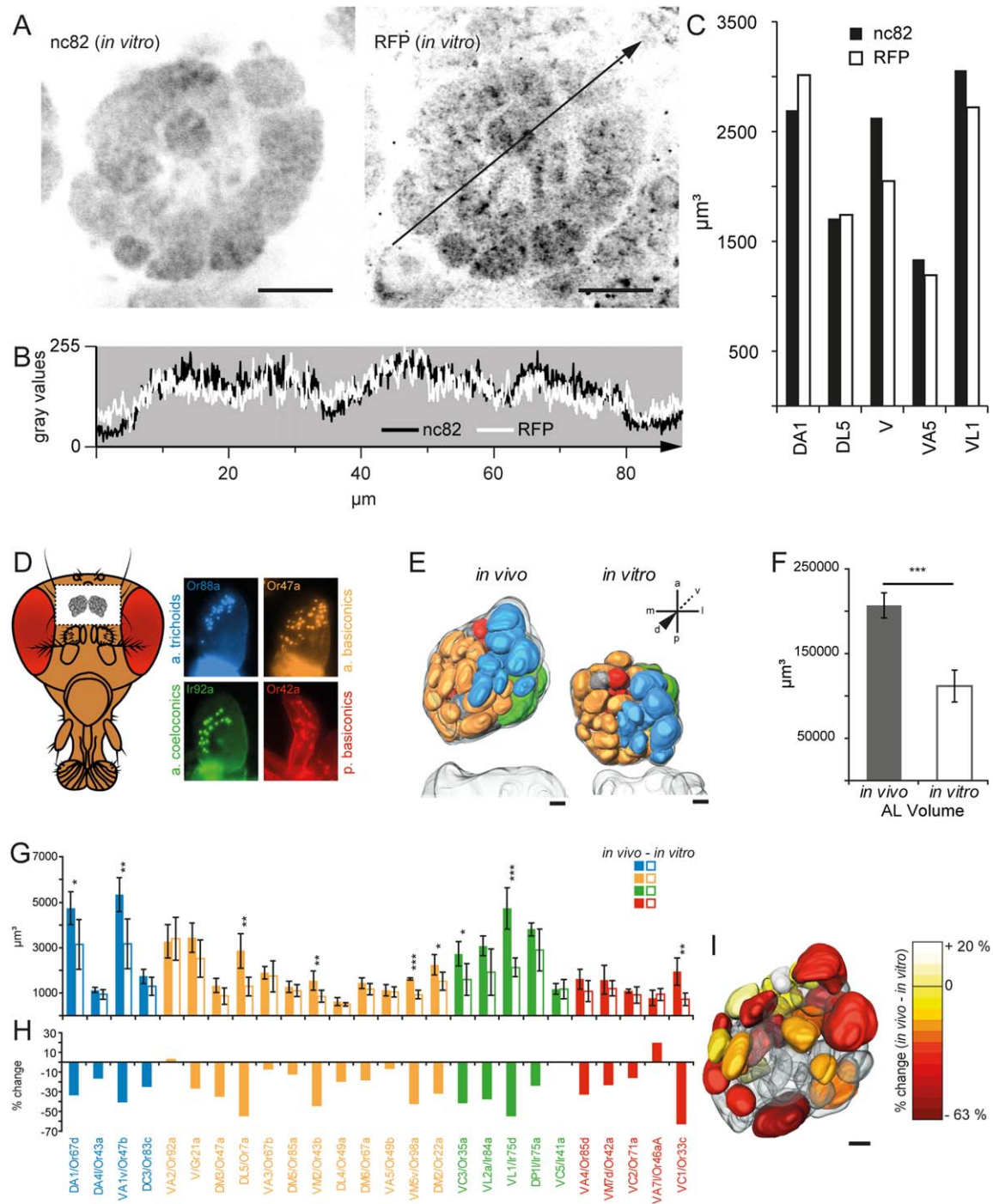


Figure 1. Characterization of volumetric differences between *in vivo* and *in vitro* antennal lobes. **A:** Immuno double staining of an *in vitro* *Drosophila* antennal lobe (AL) with nc82 (left) and anti-RFP (right) in inverted gray values. Both stainings visualize the glomerular borders in a comparable manner. **B:** Profile of gray values along a transverse line across the AL as shown in (A) of the nc82 (black line) and anti-RFP (white line) double-stained AL. **C:** Volume comparison of a subset of five glomeruli between the nc82 (black) and anti-RFP (white) staining in one animal. **D:** Left, schematic of the head of *D. melanogaster* with a superimposed window depicting the *in vivo* dissection area and position of the ALs. Right, images of the olfactory organs showing the distribution of four sensillum types representative for the different classes of sensilla acquired with specific GAL4 receptor lines (blue: antennal trichoids (Or88a-GAL4 / type: At4), yellow: antennal basiconics (Or47a-GAL4 / type: Ab5), green: antennal coeloconics (Ir92a-GAL4 / type: Ac1), red: palp basiconics (Or42a-GAL4 / type: Pb1)). **E:** Dorsal view on two 3D-reconstructed ALs with individual glomeruli. The left AL derives from an *in vivo* scan of the END1-2 neuropil labeling; the one on the right represents an AL after *in vitro* processing and nc82 antibody staining. The color code denotes the glomerular innervation by the four sensilla classes as shown in (D). **F:** Comparison of the averaged whole AL volume *in vivo* ($n = 4$) and *in vitro* ($n = 7$) based on 3D-reconstructions as shown in (E). Error bars represent SD; *** $P < 0.001$, unpaired *t*-test. **G:** Comparison of the glomerulus-specific volume of *in vivo* (filled columns, $n = 4$) and *in vitro* processed ALs (empty columns, $n = 7$) based on 3D-reconstructions as shown in (E). Error bars represent SD; * $P < 0.5$, ** $P < 0.1$, *** $P < 0.001$, unpaired *t*-test. The color code indicates the classes of sensilla as shown in (D). **H:** Percentage of the glomerulus-specific volume change between the *in vivo* and *in vitro* datasets indicating an unequal fixation effect. **I:** Glomerulus-specific volume changes are heat-mapped onto the reconstructed *in vivo* AL. White represents an increase, yellow no change, and red a decrease in volume. For glomerular identities please see Fig. 5E. Scale bars = 20 μm .

and embedded in VectaShield (Vector Laboratories, Burlingame, CA) between a 22 × 22 mm and a 24 × 60 mm object slide. Scans were carried out with an Axio Imager.Z1 (Carl Zeiss, Jena, Germany) using a 20× water immersion objective (W Plan-Apochromat 20×/0.8, Carl Zeiss) from both sides of each antenna. The *in vivo* fly brain dissection was carried out as previously described (Strutz et al., 2012) in saline (130 mM NaCl, 5 mM KCl, 2 mM MgCl₂, 2 mM CaCl₂, 36 mM saccharose, and 5 mM Hepes) at pH 7.3 adjusted with 1 M NaOH.

In vitro immunohistochemistry

Female flies aged between 3 to 6 days after hatching were used. Fly brains were dissected and stained as described by Wu and Luo (2006) and scanned with a Zeiss LSM 710 NLO confocal microscope (Carl Zeiss, Jena, Germany) set on a Smart Table UT2 (Newport, Irvine, CA) using a 40× water immersion objective (W Plan-Apochromat 40×/1.0 DIC M27; Carl Zeiss). Primary antibodies were rabbit α-RFP (1:100, Invitrogen, Eugene, OR; Molecular Probes Cat. no. R10367 RRID:AB_2315269) against DsRed and mouse α-nc82 (1:30; Developmental Studies Hybridoma Bank [DSHB], Iowa City, IA; Cat. no. nc82 RRID:AB_528108). All primary antibodies are characterized below. Secondary antibodies were Alexa Fluor 633, goat anti-rabbit IgG (Invitrogen, Eugene, OR, Life Technologies; Cat. no. A21070 RRID:AB_10562894) and Alexa Fluor 488, goat anti-mouse IgG (both 1:250, Invitrogen, Life Technologies Cat. no. A11001 RRID:AB_10566289) (Fig. 1A–C). For the generation of the *in vitro* AL atlas (Fig. 1E), primary antibody was mouse α-nc82 (1:30; DSHB), secondary antibody was Alexa Fluor 488, goat anti-mouse IgG (1:200, Invitrogen). Z-stacks were scanned at 1-μm intervals at 512 × 512 or 1024 × 1024 pixel resolution.

We used monoclonal nc82 raised in a hybridoma screen against fly head homogenate. It was characterized by Daniels et al. (2008) with respect to labeling neuropil compartments in larval and adult *Drosophila*. Polyclonal RFP antibody was raised against full-length recombinant denatured and nondenatured TagRFP and characterized by Marcucci et al. (2011). No immunolabeling was observed with wildtype fly brains (data not shown).

3D reconstruction and identification of glomeruli

Confocal scans were obtained with a ZEISS LSM 710 NLO using a 40× water immersion objective (W Plan-Apochromat 40×/1.0 DIC M27; Carl Zeiss). Individual glomeruli were reconstructed using the segmentation

software AMIRA 5.5.0 (FEI Visualization Sciences Group, Burlington, MA; Advanced 3D Visualization and Volume Modeling, RRID:nif-0000-00262). The identification of glomeruli was verified by *in vivo* scans using 34 specific receptor GAL4-lines crossed to UAS-GCaMP3.0 and combined with the END1-2 neuropil labeling. We analyzed scans of at least three specimens for each GAL4 receptor line. Using information on the voxel size from the LSM scans as well as the number of voxels labeled for each neuropil, we calculated the volume of the glomeruli and the whole AL.

We reconstructed *in vivo* ALs including all glomeruli of four female flies and selected one representative AL on the basis of the best signal-to-noise ratio, i.e., the clearest END1-2 staining, as the template for the *in vivo* AL atlas. The other specimens were used as references. All 54 glomeruli that are included in the *in vivo* atlas could be identified unambiguously in all reference specimens. Although our *in vivo* AL atlas derives from a female AL, it can be used as a template for male flies as well, since it has been shown that the glomerular identity is not clearly sexually dimorphic (Laisue et al., 1999). The sexual dimorphism that has been observed so far affects the volume of the three glomeruli DA1, VA1v, and VL2a (Kondoh et al., 2003; Stockinger et al., 2005). These glomeruli are innervated by OSNs that express the male-specific fruitless gene and are larger in males than in females. However, since these studies derive from *in vitro* brain dissections, it still remains to be elucidated whether the described dimorphism holds true for the *in vivo* condition or whether more striking effects will be observed.

Generation of a 3D PDF of the *in vivo* AL atlas followed the described procedure (Ruthensteiner and Heß, 2008; Rybak et al., 2010). Reconstructed surfaces from AMIRA were imported to FIJI (ImageJ 1.48r, National Institutes of Health, Bethesda, MD; FIJI, RRID:SciRes_000137) and transformed into wavefront format (.obj). Sorting and grouping of the glomerular materials was done in Deep Exploration (5.0.5, Right Hemisphere, San Ramon, CA) saved in u3d-format. Final adjustments of visualization parameters were done in Adobe Acrobat X Pro (Adobe Systems, San Jose, CA).

Statistical analyses

Central coordinates (xyz) for each glomerulus were extracted from the reconstructed specimens' 3D labels via the "MaterialStatistics" tool in AMIRA. Reconstructed ALs were aligned beforehand by inverting corresponding axes, turning all ALs into right ones facing the same direction. Furthermore, distribution along the x-, y-, and z-axes was aligned, i.e., all *in vivo* and *in vitro* ALs were registered in a Cartesian coordinate

system of the same dimensions. The single axes are oriented in mediolateral (x), anteroposterior (y), and dorso-ventral (z) directions through the AL. We calculated the minimal and maximal glomerular central coordinates along each axis for each individual AL as well as the average of these minimal and maximal coordinates across all *in vivo* and *in vitro* ALs separately. The difference between the individual and average minimal and maximal values was averaged per specimen, displaying the necessary shift along all three axes for every glomerulus per specimen to align the entire AL. This was done by subtracting the average per axis per specimen from each of its central glomerular coordinates. The same procedure was carried out for the averaged *in vivo* and *in vitro* ALs. Distances between central glomeruli coordinates were calculated as Euclidean distances in the Cartesian coordinate system for the *in vivo* and *in vitro* averaged ALs:

$$ED = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

To eliminate the impact of the overall shrinkage of the ALs on the relative change, distances were normalized to the maximum distance in the average *in vivo* and *in vitro* AL datasets, respectively. To calculate the position change between a pair of glomeruli from *in vivo* to *in vitro* we placed one of the two glomeruli in the center of the coordinate system ($glom_0$) by subtracting its xyz coordinates from all other glomeruli in all specimens of the *in vivo* and *in vitro* dataset. Then we calculated the cosine of the angle of the position vector between $glom_0$ and another glomerulus *in vivo* ($coord_1$) and *in vitro* ($coord_2$). Based on the cosine we calculated the radian measure of the angle using the arc cosine and multiplied the radian with $180/\pi$, converting it to degree. All analyses were done in Excel.

$$\cos \varphi = \frac{coord_1 * coord_2}{|coord_1| * |coord_2|}$$

$$\cos \varphi = \frac{(x_1 * x_2 + y_1 * y_2 + z_1 * z_2)}{\sqrt{x_1^2 + y_1^2 + z_1^2} * \sqrt{x_2^2 + y_2^2 + z_2^2}}$$

RESULTS

In vivo neuropil labeling

We generated a transgenic fly line that expresses the presynaptic protein *n*-synaptobrevin directly fused to the red fluorescent protein DsRed under the control of the pan-neuronal promotor *elav* (Yao et al., 1993), subsequently called END1-2. Our *in vivo* END1-2 labeling features a comparable neuropil staining as the most commonly used neuropil-specific monoclonal antibody nc82 *in vitro* (Hofbauer, 1991; Laissue et al., 1999; Rein et al., 2002). In order to show that the END1-2

neuropil labeling correlates with the nc82 antibody staining, we carried out a double *in vitro* immunostaining with anti-RFP against DsRed and nc82 (Fig. 1A). Both antibodies show a distinct but comparable staining of glomeruli. To quantify the distribution of tagged antibodies, we analyzed gray values across the AL. Although the stains are not completely overlapping, in their gray value profiles similar glomerular borders can be seen (Fig. 1B). In addition, staining nc82 and anti-RFP results in similar glomerular volumes as analyzed for a subset of five glomeruli (Fig. 1C).

We next verified that the protein DsRed itself is not influencing the glomerular volume *in vivo*, since it has been shown to oligomerize (Baird et al., 2000). In order to visualize individual glomeruli in flies that are END1-2 negative, we selectively expressed UAS-GCaMP3.0 in Or22a-OSNs. Subsequently, we compared the volume of the respective glomerulus DM2 with and without the expression of END1-2. The results do not reveal any significant difference in volume ($n = 3$, data not shown).

Volumetric fixation effects on the antennal lobe

In order to analyze the effects caused by *in vitro* processing, we first quantified volumetric differences between the *in vivo* and *in vitro* AL morphology. To reduce the preparatory artifacts onto the brain to a minimum, we executed an *in vivo* preparation usually used for functional imaging (Silbering et al., 2012; Strutz et al., 2012). By immobilizing the fly in a Plexiglas stage and opening the head capsule under saline, we were able to remove the tracheal sacks and fat tissue to gain free access to the ALs; during this time, the brain was retained in the head capsule and the antennal nerves remained intact (Fig. 1D). Using the *n*-synaptobrevin::DsRed labeling as a neuropil marker in the live fly, we scanned the brain with the confocal microscope and reconstructed the whole AL, including all glomeruli, *in vivo* (Fig. 1E, left). In parallel, we carried out antibody staining according to Wu and Luo (2006) with the neuropil marker nc82 *in vitro* as well as scanned and reconstructed these *in vitro* ALs using the same procedures as for the *in vivo* data (Fig. 1E, right). In addition, to facilitate the comparison between the *in vivo* and *in vitro* generated datasets, we reconstructed the mushroom body β/β' - and γ lobes as anatomical landmarks. As a quantitative measure, we used the number of voxels in the reconstructed ALs and calculated the total AL volume based on their dimensions (Fig. 1F). A comparison of the *in vivo* and *in vitro* datasets shows that the AL volume decreased by about 43% due to the dissection and fixation procedure. The shrinkage of the whole AL neuropil *in vitro*

would not impair the identification of glomeruli for *in vivo* experiments, if the glomeruli were evenly affected. We therefore quantified and compared the volume of individual glomeruli for the *in vivo* and *in vitro* dataset.

Glomerulus-specific analysis of volumetric differences between *in vivo* and *in vitro* brains

In order to ensure correct glomerulus identification, the END1-2 flies were recombined with UAS-GCaMP3.0 and crossed with 34 receptor GAL4-lines each, i.e., two-thirds of glomeruli were thus labeled according to their corresponding OSN innervation (Couto et al., 2005; Fishilevich and Vosshall, 2005; Silbering et al., 2011). For the detailed quantification of the fixation artifacts, we selected a group of 26 representative glomeruli spanning throughout the AL neuropil and covering different classes of sensilla. Notably, we observed a clear unequal reduction of these glomerular volumes due to fixation (Fig. 1G). Whereas some glomeruli revealed a volume reduction of up to 60% (e.g., VC1) due to *in vitro* processing, other glomeruli shrank by 30% (e.g., DA1) or retained their volume (e.g., VA2) (Fig. 1H). By mapping these glomerulus-specific volumetric effects onto the reconstructed *in vivo* AL, it is obvious that the shrinkage correlates neither with the specific position nor with the total volume of the individual glomeruli (Fig. 1I). Since every glomerulus is shrinking to a different degree, in the next step we analyzed whether the *in vitro* processing caused any geometrical modifications of the glomerular arrangement.

Effect of fixation on the AL geometry

To evaluate possible dislocations of glomeruli following *in vitro* processing (Fig. 2A), we determined Euclidean distances by giving each glomerulus an x, y, and z-coordinate representing its center in the Cartesian system (see Materials and Methods). For each glomerulus, center-to-center distances were calculated in relation to every other glomerulus and subsequently averaged per glomerulus over the *in vivo* and *in vitro* datasets. We then normalized the Euclidean distances to the maximum glomerular distance per dataset and subtracted the normalized *in vivo* distance from the *in vitro* distance. Hence, a negative Δ Distance represents the approach of two glomeruli, while a positive Δ Distance reveals the increased interglomerular distance due to *in vitro* processing (Fig. 2B). Averaging the positive and negative distance changes per glomerulus over all *in vivo* and *in vitro* specimens, respectively (Fig. 2C, $n = 4-7$), shows various kinds of strong glomerular dislocations; for example, glomeruli VA2 and VM2 increase

their distance to most other glomeruli in the *in vitro* dataset, while DL5 and DP11 show the opposite effect. Glomerulus VA1v is affected in both directions, as it ranks highest for an increased as well as decreased interglomerular distance change. Mapping these data onto the reconstructed *in vivo* AL illustrates the spatial distribution of the various effects of glomerular dislocation for the whole AL (Fig. 2D,E). One major reason for the strong geometrical modifications between the *in vivo* and *in vitro* AL is most likely the transection of the antennal nerve, a process in which the brain is extracted from the head capsule. Since the nerve provides tension to the AL, severing it deforms the whole AL neuropil (Fig. 1E).

Even though some glomeruli do not reveal pronounced interglomerular dislocation effects (e.g., DA4I, DM5, and VM7d), we cannot exclude that their position is affected without substantially changing their interglomerular distance. To estimate glomerular distortions independent of their distance, we determined the angle between the two position vectors of each possible glomerular combination for the *in vivo* and *in vitro* datasets (Fig. 2F,G). In order to obtain a glomerulus-specific estimation of the overall angle between *in vivo* and *in vitro* vectors, we calculated the average angle per glomerulus (Fig. 2H). Most glomeruli exhibit an angle above 30° , meaning that the positional arrangement of most glomeruli is considerably affected by the *in vitro* processing. Again, mapping these glomerular distortions on the reconstructed AL (Fig. 2I) reveals that the posterior area is affected strongest. In summary, due to the strong and so far unpredictable fixation artifacts in AL volume and geometry, a digital *in vivo* 3D atlas of the fly AL is essential in order to identify individual glomeruli in *in vivo* studies.

In vivo atlas of the antennal lobe

Previous *in vitro* studies established several AL atlases for *Drosophila* with varying nomenclature and numbers of glomeruli (Stocker et al., 1983; Laissue et al., 1999; Couto et al., 2005; Endo et al., 2007; Silbering et al., 2011). We are referring to the most recent nomenclature from Tanaka et al. (2012), who describe glomerulus VM7 as VM7d and the former unnamed glomerulus "1" (Couto et al., 2005) as its ventral counterpart VM7v. The split of glomerulus VC3 into VC3l and VC3m could not be confirmed at a morphological level, since both parts are innervated by OSNs expressing Or35a (Silbering et al., 2011). We therefore refer to this glomerulus as VC3. Even though no OR is yet known for glomerulus VA7m, we kept the anatomical division in the lateral and medial parts, since the two parts represent different subunits defined by the

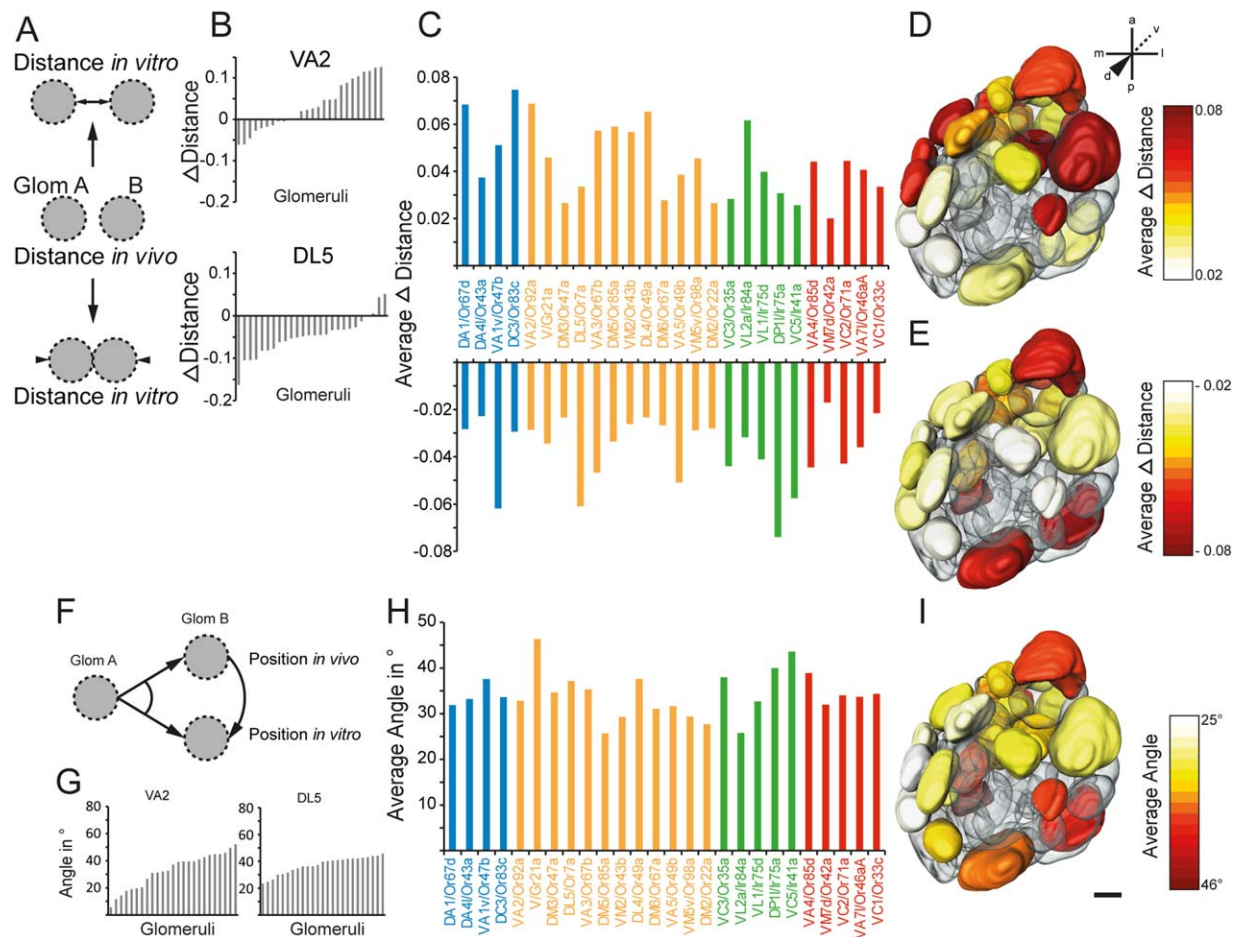


Figure 2. *In vitro* dissection and fixation leads to geometrical modifications of various glomeruli. **A:** Schematic of the investigated Euclidean distance changes between glomeruli as shown in B–E (for details, see Materials and Methods). **B:** Example of normalized center-to-center distance changes (Δ Distance) of two representative glomeruli in relation to all other glomeruli between the *in vivo* and *in vitro* dataset. **C:** Averaged positive (top) and negative (bottom) normalized distance changes per glomerulus in relation to all other glomeruli between all *in vivo* ($n = 4$) and *in vitro* specimens ($n = 7$). Positive values indicate an increased interglomerular distance; negative values represent a decrease. Various glomeruli display strong positive and/or negative distance changes underlining nonuniform dislocations of the glomerular arrangement. The color code indicates the classes of sensilla as shown in Figure 1. **D,E:** Positive (top) and negative (bottom) distance changes are heat-mapped onto the reconstructed *in vivo* AL. **F:** Schematic of the investigated angle between two positional vectors of each glomerular pair as shown in (G–I). **G:** Example of calculated angles of two representative glomeruli between their positional vectors in relation to all other glomeruli of the *in vivo* and *in vitro* datasets. **H:** Averaged angles between the *in vivo* ($n = 4$) and *in vitro* ($n = 7$) positional vectors for each glomerulus. The positional arrangement of most glomeruli is considerably affected by dissection and *in vitro* processing. **I:** Averaged angles between *in vivo* and *in vitro* positional vectors are heat-mapped onto the reconstructed AL. For glomerular identities please see Fig. 5E. Scale bar = 20 μ m.

convergence of OSNs expressing Or46aA to the VA71 part (Couto et al., 2005). Moreover, we fused glomerulus VP1 with glomerulus VM6, since they are morphologically not distinguishable and both are said to share most of their projection neurons (Yu et al., 2010), indicating their uniform nature. In addition, our *in vivo* AL atlas comprises the often neglected glomeruli VP1 and VP4 (Silbering et al., 2011; Kain et al., 2013), although their exact morphology is cryptic without the simultaneous expression of Ir40a-GAL4 that we employed for reliable identification. We also included the thermosensory

glomeruli VP2 and VP3 in our AL atlas; these appear as part of the AL rather than as part of the proximal antennal protocerebrum as described previously (Gallio et al., 2011), since their input neurons are ascending along with the OSNs of other glomeruli.

By selectively expressing 34 receptor GAL4-lines combined with GCaMP3.0 into the END1-2 background (see Materials and Methods), we generated a new digital 3D atlas of the *in vivo* *Drosophila* AL (Fig. 3). First, we selected the most suitable confocal stack out of 13 scans of the END1-2 neuropil labeling (Fig. 3A) as a

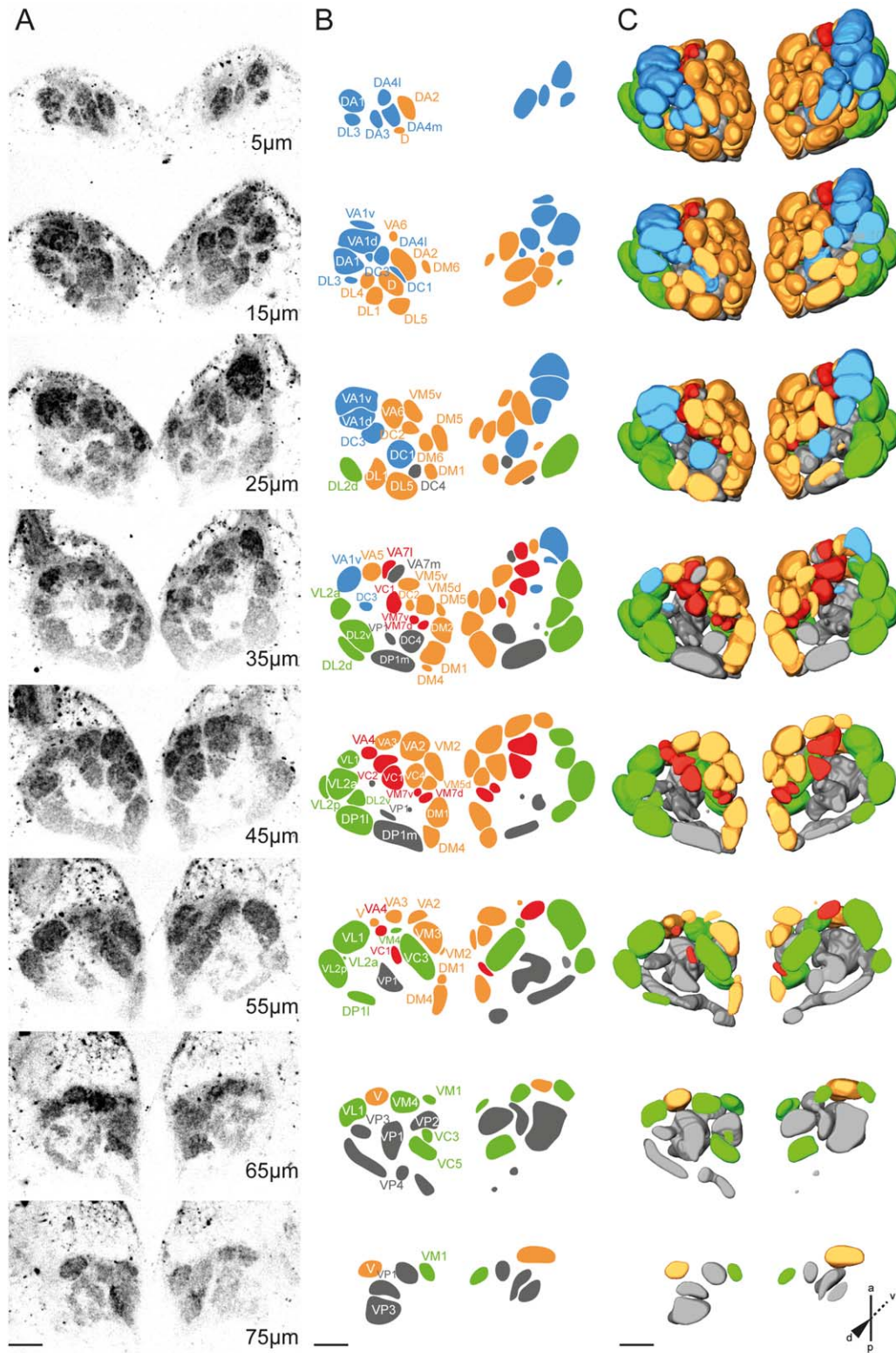


Figure 3. *In vivo* 3D atlas of the *Drosophila* antennal lobe. **A:** Representative confocal stack of an *in vivo* AL expressing the END1-2 neuropil labeling. Eight planes from dorsal to ventral (top to bottom) through a female AL are shown at 10-μm intervals displayed in an inverted gray scale. **B:** Identified and reconstructed glomeruli of the confocal stack shown in (A). **C:** Dorsal view on the 3D-reconstruction of the labels shown in (B). The glomeruli are successively removed as the scan moves from dorsal to ventral through the AL. The color code indicates the classes of sensilla as shown in Figure 1, extended by glomeruli that receive input from the grooved coeloconic sensilla of the sacculus and the arista shaft (shown in gray). Scale bars = 20 μm.

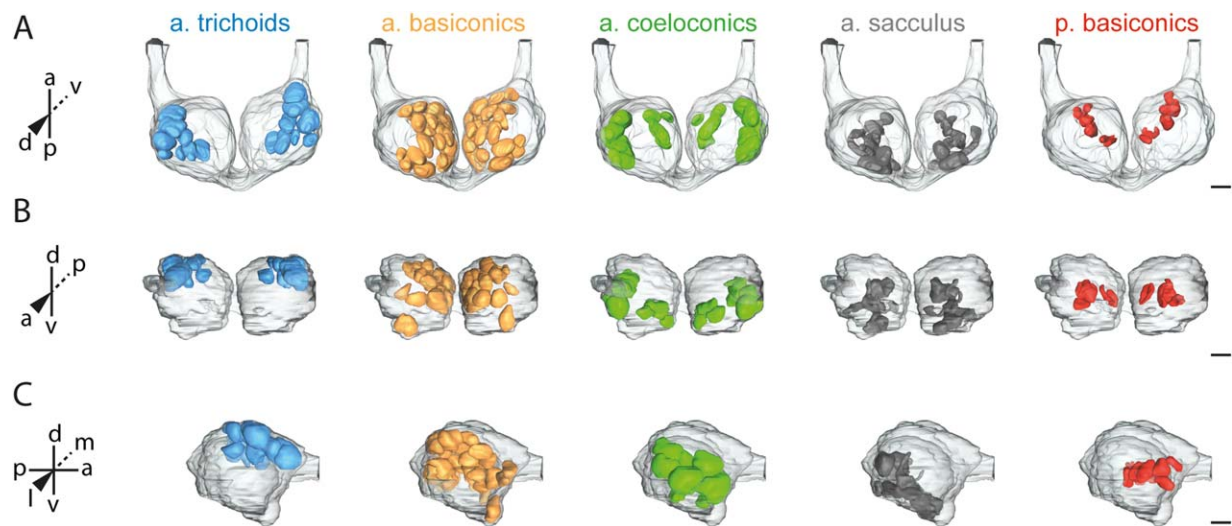


Figure 4. Segregation of the classes of sensilla in the AL. **A–C:** Innervation of glomeruli separately represented for each class of sensillum of the corresponding OSN class in a dorsal (A), frontal (B), and lateral (C) view. The color code refers to Fig. 3. For glomerular identities please see Figure 5E–G. Scale bars = 20 μ m.

template for 3D reconstructions, based on the best signal-to-noise ratio throughout the entire scanned distance of ~ 100 μ m. To securely identify the reconstructed glomeruli in the selected stack, in the initial scans we employed specific receptor GAL4-lines for 63% of the AL glomeruli and used the stereotypical OSN innervation patterns and their positions to verify glomerular identity. Based on the precise identification via the receptor GAL4-lines, we morphologically identified the remaining neuropils comprising a total of 54 glomeruli (Fig. 3B). The template scan was then properly reconstructed, improved through the OSN mapping and morphological assignment, and color-coded according to the class of sensilla projecting onto each glomerulus (Fig. 3C). In addition, we performed complete 3D glomerular reconstructions of three further specimens and used them as references. All 54 glomeruli that are included in the *in vivo* AL atlas could be identified unambiguously in all of these reference specimens.

Application of the *in vivo* AL atlas

Our *in vivo* AL atlas represents an accurate match for *in vivo* applications and provides a novel basis for reconsidering neuroanatomical features as exemplified by the glomerular clustering according to the class of sensilla of the OSN input (Couto et al., 2005). The discovery of ionotropic glutamate receptors in insects (IRs) (Benton et al., 2009) underlined this functional and morphological subdivision of the AL and revealed a clear segregation of glomeruli innervated by olfactory receptors (ORs) and IRs (Silbering et al., 2011).

Revisiting this feature for the *in vivo* condition, the above-described segregation is present also in the live AL (Fig. 4). In addition, the glomerulus cluster that receives input from coeloconic sensilla can be further divided into two subtypes: Glomeruli that are innervated by OSNs from the sacculus expressing Ir40a and Ir64a (Ai et al., 2010; Silbering et al., 2011) are located separately from glomeruli receiving input from the remaining coeloconic sensilla. This segregation is most obvious in the lateral view of the AL (Fig. 4C), where glomeruli originating in the sacculus and situated most ventroposteriorly are followed in dorsoanterior direction by three layers of glomeruli receiving input from coeloconic, basiconic (antenna and palp), and trichoid sensilla.

As another example, we demonstrate the application of two of the most frequently used GAL4-lines in olfactory research: Orco-GAL4 and GH146-GAL4. The Orco-GAL4 line labels all OSNs that express the coreceptor Orco, formerly known as Or83b (Larsson et al., 2004); the enhancer trap line GH146-GAL4 labels the majority of uniglomerular projection neurons (PNs) in the AL (Stocker et al., 1997). Confocal *in vivo* scans of the Orco-GAL4 line combined with END1-2 and GCaMP3.0 reveal the clear innervation of the majority of glomeruli (39 glomeruli out of 54); the IR-positive glomeruli and the thermosensory center consisting of VP2 and VP3 are not labeled (Fig. 5A,B). The Orco-negative glomeruli group at the ventroposterior region of the AL, as they are congruent with the glomeruli innervated by coeloconic sensilla. Similarly, the innervation of the GH146-GAL4 line labels a total of 39 glomeruli; however, only

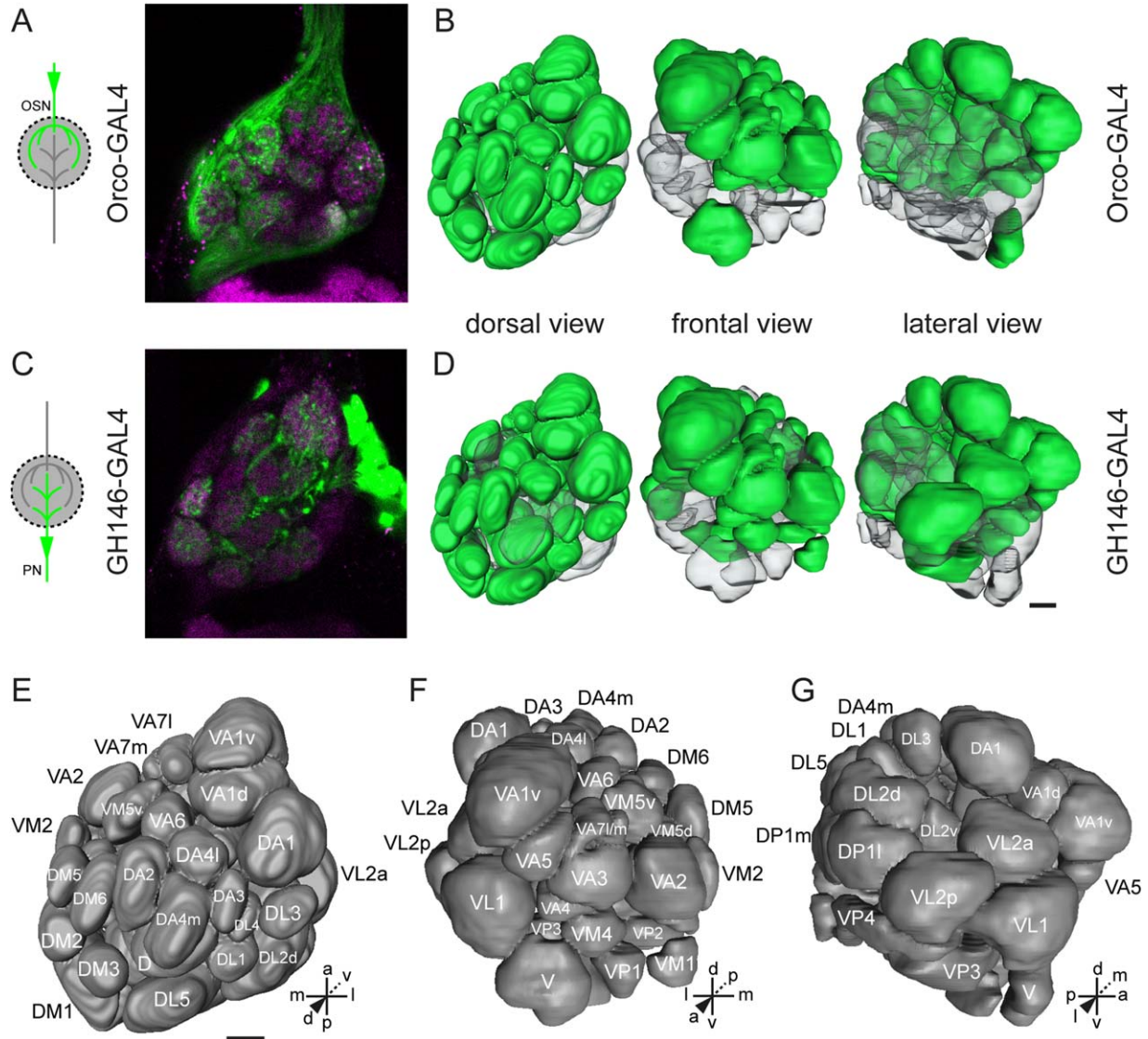


Figure 5. Characterization of the glomerular innervation patterns of two widely used GAL4-lines. **A:** Representative *in vivo* confocal scan of an AL expressing GCaMP3.0 under the control of the Orco promoter (green) combined with the END1–2 neuropil labeling (magenta) at 20 μ m depth. The Orco-GAL4 line labels the majority of olfactory sensory neurons as depicted in the schematic on the left side. **B:** Glomerular innervations of the Orco-GAL4 line represented via the *in vivo* 3D AL atlas (green: innervated; transparent: not innervated). Depicted are the dorsal, frontal, and lateral views onto the AL. **C:** Representative *in vivo* confocal scan of an AL expressing GCaMP3.0 under the control of the GH146-GAL4 driver line (green) combined with the END1–2 neuropil labeling (magenta) at 30 μ m depth. The GH146-GAL4 line labels the majority of projection neurons as depicted in the schematic on the left side. **D:** Glomerular innervations of the GH146-GAL4 line represented via the *in vivo* 3D AL atlas (green: innervated; transparent: not innervated). Depicted are the dorsal, frontal, and lateral views onto the AL. **E–G:** Complete glomerular assignment of the *in vivo* AL atlas from dorsal (E), frontal (F), and lateral views (G) corresponding to the perspectives in (B) and (D). Scale bars = 20 μ m.

32 of which overlap the innervation of Orco-GAL4 (Fig. 5C,D; for glomerular identities, see Fig. 5E–G). Hence, if these two lines are employed simultaneously, functional studies that compare the input and output properties of specific glomeruli should be analyzed with caution. Although GH146-GAL4 is said to label one ventral PN innervating the whole AL (Marin et al., 2002), we were not able to observe any multiglomerular innervation.

Due to the sparse glomerular innervation of multiglomerular PNs (Lai et al., 2008), the GCaMP3.0 expression level in this PN, which is below the detection level, can also be ignored for imaging experiments of the whole AL. For a detailed overview, all innervated glomeruli are listed for each GAL4-line in Table 1.

In summary, our *in vivo* AL atlas can be used to reliably identify glomeruli throughout the entire neuropil in

TABLE 1.
Glomeruli Innervated by the Orco-GAL4 and
GH146-GAL4 Line

Glomerulus	Receptor type	Sensillum type	Orco- GAL4	GH146- GAL4
D	Or69a	Ab9		
DA1	Or67d	At1		
DA2	Or56a	Ab4		
DA3	Or23a	At2		
DA4l	Or43a	At3		
DA4m	Or2a	At3		
DC1	Or19a	At3		
DC2	Or13a	Ab6		
DC3	Or83c	At2		
DC4	Ir64a	Sac III		
DL1	Or10a	Ab1		
DL2d	Ir75a/b/c	Ac3		
DL2v	Ir75a/b/c	Ac3		
DL3	Or65a	At4		
DL4	Or49a(Or85f)	Ab10		
DL5	Or7a	Ab4		
DM1	Or42b	Ab1		
DM2	Or22a	Ab3		
DM3	Or47a	Ab5		
DM4	Or59b	Ab2		
DM5	Or85a(Or33b)	Ab2		
DM6	Or67a	Ab10		
DP1l	Ir75a	Ac2		
DP1m	Ir64a	Sac III		
V	Gr21a	Ab1		
VA1d	Or88a	At4		
VA1v	Or47b	At4		
VA2	Or92a	Ab1		
VA3	Or67b	Ab9		
VA4	Or85d	Pb3		
VA5	Or49b	Ab6		
VA6	Or82a	Ab5		
VA7l	Or46aA	Pb2		
VA7m	?	?		
VC1	Or33c	Pb2		
VC2	Or71a	Pb1		
VC3	Or35a	Ac3		
VC4	Or67c	Ab7		
VC5	Ir41a	Ac2		
VL1	Ir75d	Ac1/2/4		
VL2a	Ir84a	Ac4		
VL2p	Ir31a	Ac1		
VM1	Ir92a	Ac1		
VM2	Or43b	Ab8		
VM3	Or9a	Ab8		
VM4	Ir76a	Ac4		
VM5d	Or85b	Ab3		
VM5v	Or98a	Ab7		
VM7d	Or42a	Pb1		
VM7v	Or59c	Pb3		
VP1	Ir40a	Sac I		
VP2	hot	Arista		
VP3	cold	Arista		
VP4	Ir40a	Sac I		

These two lines refer to the olfactory sensory neurons and projection neurons, respectively. Black filled cells indicate innervation by the indicated line; white represents no innervation. Ab, antennal basiconic; Ac, antennal coeloconic; At, antennal trichoid; Pb, palp basiconic; Sac, sacculus; Or, olfactory receptor; Ir, ionotropic receptor; Gr, gustatory receptor.

the live fly brain; the atlas preserves the spatial arrangement of individual glomeruli as well as interglomerular differences regarding shape and size.

DISCUSSION

In vivo atlas for enhanced applicability

In this study we used a newly generated transgenic fly line, referred to as the END1-2 line, to provide *in vivo* neuropil labeling of the AL; this label is comparable to the nc82 immunostaining of *in vitro* whole mount brains (Fig. 1A-C) (Hofbauer, 1991; Laissue et al., 1999; Rein et al., 2002). The END1-2 line enables synaptically dense glomerular areas to be separated from synaptically sparse interglomerular areas (Boeckh et al., 1970; Tolbert and Hildebrand, 1981) by the expression of the presynaptic fusion protein *n*-synaptobrevin::DsRed in all neurons. By obtaining confocal stacks of this line and reconstructing individual glomeruli, we generated a 3D map of the AL of *Drosophila melanogaster* that accurately matches the AL in the living fly (Fig. 3). By taking advantage of the near-complete maps of receptor gene expression and OSN targeting (Couto et al., 2005; Fishilevich and Vosshall, 2005; Silbering et al., 2011), we verified the identity of 63% of AL glomeruli using specific receptor-GAL4 lines. This assisted identification allowed us to develop an *in vivo* AL atlas and a way to label neuropils that enables the orientation within and attribution of individual brain regions without *in vitro* processing. In addition, we demonstrate two applications of the new *in vivo* AL atlas by revisiting the segregation of glomeruli innervated by different types of sensilla (Fig. 4) (Couto et al., 2005), as well as by characterizing the glomerular innervation patterns of two of the most crucial GAL4-lines for targeting OSNs and PNs in the fly AL (Fig. 5).

In light of the number of glomeruli published over the last few years, which ranges from 35 (Stocker et al., 1990) to over 43 (Laissue et al., 1999) and up to 56 in current nomenclature (Tanaka et al., 2012), clearly the identification of glomeruli is not trivial. The subdividing of glomeruli, such as DL2, DP1, VA1, VA7, VC3, VL2 (Laissue et al., 1999), DA4, VM5, and VM7 (Couto et al., 2005) and the merging of others, such as VA1l and VA1m, VC3l and VC3m (Couto et al., 2005) as well as the addition of newly defined ones (e.g., VC5; Silbering et al., 2011) outline the challenge of distinguishing reliably between glomeruli. Consider glomerulus VM6, which cannot be distinguished from glomerulus VP1 by plain morphology. We merged the two glomeruli as VP1 since Yu et al. (2010) only observed PNs innervating both structures together via a twin-spot MARCM study. In addition, we were not able to assign glomerulus DL6,

which has been mentioned in two studies (Marin et al., 2005; Yu et al., 2010). However, its limited description to date leads us to assume that this glomerulus has been mislabeled and represents in fact glomerulus DC3. Hence, the new *in vivo* AL atlas of the vinegar fly consists of 54 glomeruli, of which 52 are innervated by chemosensory and two by thermosensory neurons (Galio et al., 2011).

In vivo* compared to *in vitro

Our idea to create an *in vivo* AL atlas derived from the need to identify glomeruli in functional imaging experiments. Glomerular identification has proven challenging for *in vivo* data with the available *in vitro* atlases. Moreover, since there are not many investigations into the fixation artifacts of flexible neuropils such as the AL, we quantified the volumetric and geometrical modifications, such as shrinkage, dislocation, and distortion following *in vitro* processing. These three substantial effects of fixation on the AL were distributed dissimilarly across the AL neuropil (Figs. 1, 2). First, the heterogeneous shrinkage of glomeruli is most likely due to the different density of intraglomerular arborizations of OSNs, PNs, or LNs (Ignell et al., 2005), which leads to a small volumetric decrease in densely packed glomeruli, while sparsely innervated glomeruli are strongly affected.

Second, the dislocation of glomeruli during fixation affects mainly those glomeruli located anteriorly (VA1v, DA1, VA2) and posteriorly (DL5, DP1l), which reflects a longitudinal distortion of the entire AL neuropil. This global deformation is most likely based on the transection of the antennal nerve, which is required for *in vitro* dissection. In addition, the transection causes general changes in glomerular visibility: Under *in vivo* conditions the anterior glomeruli of the VA cluster are invisible when viewed from a dorsal perspective; they become visible in the surface layer following *in vitro* processing. The opposite effect happens for the posteriorly located glomeruli of the DL and DP clusters, as they are hidden beneath the DA glomeruli in the *in vitro* dissection.

Third, in addition to analyzing the dislocation of several glomeruli, we also investigated distortions regarding the relative positions of glomerular pairs. Again, we observed distortions mainly in glomerular pairs that are located anteriorly and posteriorly, which parallels the dislocation effect seen in individuals; however, the posterior glomeruli are more affected. These deformations highlight the rearrangement of most glomeruli around the ventroposterior basis, where the AL is attached to the protocerebrum. These diverse fixation artifacts emphasize the need for a thorough *in vivo* 3D AL atlas which has an enhanced application for characterizing GAL4 lines combined with the *in vivo* neuropil labeling

in single- and multiphoton imaging experiments throughout the whole AL.

***In vivo* GAL4-line evaluations**

As most of the ubiquitous GAL4-lines used for functional imaging studies are characterized by *in vitro* whole-mount brain immunochemistry, their maps suffer from the same issues as the initial identification of glomeruli described above. One can now screen any GAL4 line with the END1-2 neuropil label combined with mCD8-GFP (Lee and Luo, 1999) by maintaining the same conditions as in the actual *in vivo* experiment. END1-2, in contrast to ENG3 (Estes et al., 2000), can also be combined with various green fluorophores (Patterson and Lippincott-Schwartz, 2002; Tian et al., 2009) to visualize the neuropil while live measurements are running. For the characterization of new GAL4-lines labeling the olfactory pathway, we recommend identifying a reliable set of landmark glomeruli (such as DA1, DL5, DM2, VA2, VL1, and VM2) that is evenly spread over the whole AL neuropil.

Advantages and limitations of the new *in vivo* atlas

Our new approach for *in vivo* morphology based on END1-2 neuropil labeling has both advantages and limitations. The new method allowed us to depict the actual volume, distance, and angle artifacts influencing the *in vitro* atlases. The *in vivo* AL atlas does not suffer from those artifacts and its application in *in vivo* experiments is thus easier, especially as it has been generated from a perspective identical to the *in vivo* dissection for functional imaging experiments (Silbering et al., 2012; Strutz et al., 2012). On the other hand, this approach is limited by the age of flies used in the END1-2 staining: By using flies of 3–6 days after hatching, we obtained a clearly stained neuropil. However, stains made with older flies revealed an increasingly indistinct distribution of the DsRed labeling, which interferes with the identification of glomeruli (data not shown). Although we are not completely sure why older flies would have this effect, we assume that it represents a disassembly of the endocytotic recycling process of SNARE complexes (Südhof, 2004; Burgalossi et al., 2010). Disassembling the whole complex could lead to enriched levels of DsRed in the synaptic cleft. Alternatively, the fusion of DsRed to *n*-synaptobrevin might constrain vesicle recycling. Hence, to achieve the most distinct *in vivo* neuropil labeling, the age dependency should be kept in mind.

In summary, the new *in vivo* 3D digital atlas of the *Drosophila* AL represented here provides an invaluable

research tool for fly brain morphology and functional imaging studies. Publicly available and downloadable as a 3D PDF file, the atlas allows the AL or subgroups of glomeruli to be visualized from any angle. In addition, the *in vivo* neuropil labeling can be used to characterize the morphology of novel driver and enhancer trap lines in the commonly used GAL4 (Brand and Perrimon, 1993), LexA (Lai and Lee, 2006), and Q systems (Potter et al., 2010) for broad applications in *Drosophila* neuroscience.

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CONFLICT OF INTEREST

The authors have no known or potential conflict of interest including any financial, personal, or other relationships with other people or organizations within the years of beginning the submitted work that could inappropriately influence or be perceived to influence the work.

ROLE OF AUTHORS

All authors had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: Silke Sachse, Veit Grabe, Antonia Strutz, Bill S. Hansson. Acquisition of data: Veit Grabe, Amelie Baschwitz. Analysis and interpretation of data: Veit Grabe, Silke Sachse. Drafting of the article: Veit Grabe, Silke Sachse. Obtained funding: Silke Sachse, Bill S. Hansson. Material support (generation of GAL4-lines): Antonia Strutz. Critical revision of the article for important intellectual content: all authors. Study supervision: Silke Sachse.

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